

HASSAN ALSHAL, PH.D.

Physics Researcher and Lecturer Faculty at Santa Clara University

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Santa Clara University, 500 El Camino Real, Santa Clara, CA 95053



SCAN ME

Academic Experiences

- Lecturer**
Department of Physics & Engineering Physics, Santa Clara University
Sep 2021 – Present Santa Clara, CA, USA
- Adjunct Lecturer**
Department of Chemistry & Physics, Lincoln University
Jan 2021 – Jul 2021 Oxford, PA, USA
- Teaching Assistant**
Department of Physics, University of Miami
Sep 2013 – Dec 2020 Miami, FL, USA
- Teaching Assistant**
Department of Physics, Faculty of Science, Cairo University
Jun 2011 – Oct 2012 Giza, Egypt
- Teaching Assistant**
Department of Physics, The American University in Cairo
Sep 2010 – May 2011 New Cairo, Egypt

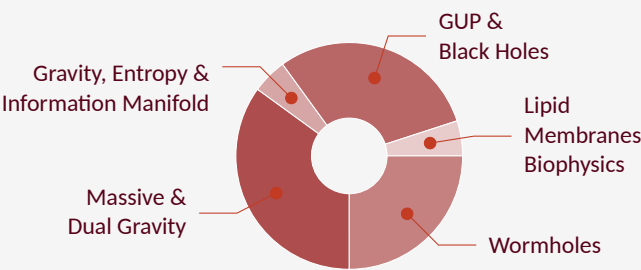
Education

- Ph.D., Physics, Dissertation Title:**
"Aspects of Massive Dual Gravity"
Supervised by: T. L. Curtright, Department of Physics, University of Miami
Sep 2013 – Aug 2020 Miami, FL, USA
- M.Sc., Physics, Thesis Title:**
"Green Functions, Sommerfeld Images, and Wormholes"
Supervised by: T. L. Curtright, Department of Physics, University of Miami
Sep 2018 – May 2019 Miami, FL, USA
- Masters of Advanced Studies (MAST), Part III of Math. Tripos**
Department of Theoretical Physics and Applied Mathematics (DAMTP), University of Cambridge
Sep 2012 – Jun 2013 (Incomplete) Cambridge, UK
- B.Sc. Physics, with Honour (Ranked 1st)**
Department of Physics, Cairo University
Sep 2006 – May 2010 Giza, Egypt
- B.Sc. Pharmaceutical Sciences**
Faculty of Pharmacy, Ain Shams University
Sep 2001 – Sep 2006 Cairo, Egypt

Awards

- Awards for Essays on Gravitation (Honorable Mention)**
Gravity Research Foundation, 2021
- Graduate Summer Research Assistantship Award**
Department of Physics, University of Miami, 2014 – 2020
- BP Cambridge Scholarships for Egypt**
Cambridge Overseas Trust, University of Cambridge, 2012
- Cairo University Award for Excellence for year 2009/2010**
Cairo University, 2010
- Schlumberger Awards for Outstanding Achievements (twice)**
Highest GPA in physics classes, Cairo University, 2008 & 2009

Research Areas



Software Skills

- Wolfram Mathematica** ●●●●●●
- Maple (with GRTensor)** ●●●●●●
- L^AT_EX (with TeXmaker)** ●●●●●●
- LINUX (with BASH)** ●●●●●●
- PYTHON (with SciPy)** ●●●●●●
- Machine Learning (Scikit)** ●●●●●●
- Deep Learning (Pytorch)** ●●●●●●
- MD Simulation (Gromacs)** ●●●●●●

Conferences & Seminars

 Undergraduates Seminars of SCU Department of Physics
Title: Quantum Computation & Machine Learning

 Department of Physics, Santa Clara University

 Jan-March 2026  Santa Clara, CA, USA

 Undergraduates Seminars of SCU Department of Physics
Title: General Relativity

 Department of Physics, Santa Clara University

 April-June 2024  Santa Clara, CA, USA

 SCU Department of Physics Seminars
Title: Massive Gravity

 Department of Physics, Santa Clara University

 Jan 2024  Santa Clara, CA, USA

 Graduate Seminars of Department of Physics
Title: Aspects of S-Duality and Curtright Fields

 Department of Physics, University of Miami

 Jan 2020  Miami, FL, USA

 Annual Coral Gables Conference
Title: Massive Dual Spin-2 Revisited

 Department of Physics, University of Miami

 Dec 2019  Fort Lauderdale, FL, USA

 Graduate Seminars of Department of Physics
Title: Green Functions Electrostatics and Wormhole Geometry

 Department of Physics, University of Miami

 Mar 2019  Miami, FL, USA

 Graduate Seminars of Department of Physics
Title: Generalized Uncertainty Principle and Quantum Gravity

 Department of Physics, University of Miami

 Nov 2018  Miami, FL, USA

 Graduate Seminars of Department of Physics
Title: Galileons and Black Holes

 Department of Physics, Cairo University

 Aug 2018  Giza, Egypt

 Graduate Seminars of Department of Physics
Title: Galileons as an Alternative Theory to General Relativity

 Department of Physics, University of Miami

 Apr 2016  Miami, FL, USA

 Graduate Seminars of MAST & Part III Math. Tripos
Title: Symmetries and Particle Physics

 Dep. of Applied Math. and Theoretical Physics, University of Cambridge

 2012 - 2013  Cambridge, UK

Nondegree Courses

 Introduction to Machine Learning
 Coursera, authorized by Duke University

 May 2021  Online

 Intro. to Python Programming
 Udacity

 Sep 2020  Online

 Summer School in Cosmology
 The Abdus Salam International Center for Theoretical Physics

 July 2010  Trieste, Italy

Journals Referee

 Adv. High Energy Phys.

 Int. J. Mod. Phys. A.

 Found. Phys.

 Int. J. Geom. Methods Mod. Phys.

 Int. J. Theor. Phys.

 Eur. Phys. J. Plus.

 Mod. Phys. Lett. A.

 Sci. Rep.

 Eur. Phys. J. C

Volunteering

 Moderator of Academic Sessions
for Molecular Dynamics Workshop

 The Second Students' Conference Of Pharmaceutical Studies, Ain Shams University

 Sep 2006 - Apr 2007  Cairo, Egypt

Nonscholar Experiences

 Pharmacy manager and
pharmacist by training

 Licensed by Ministry of Health and
Pharmacists Syndicate, Egypt

 Sep 2006 - Oct 2012  Cairo, Egypt

Journal Articles

- W. Chrifi, et al. (2025). “Lipid Composition Controls Temperature-Dependent Viral Immune Response in Human Bronchoalveolar Space”. In: DOI: 10.2139/ssrn.5583865. SSRN: 5583865.
- Alshal, Hassan, Elias C. Vagenas, and Thomas Van Kortryk (2025). “Spin-2 fields, Lee-Wick Ghosts, and GUP”. in: DOI: 10.48550/arXiv.2512.24144. arXiv: 2512.24144 [gr-qc].
- Alshal, Hassan, Leyang Ding, Adelina Hernandez, Leo A. Illing, and Ivar Rydstrom (2025). “Linearized stability of Harada thin-shell wormholes”. In: *Gen. Rel. Grav.* 57.1, p. 9. DOI: 10.1007/s10714-024-03344-3. arXiv: 2407.08945 [gr-qc].
- Alshal, Hassan (2023). “Einstein’s equations and the pseudo-entropy of pseudo-Riemannian information manifolds”. In: *Gen. Rel. Grav.* 55.7, p. 86. DOI: 10.1007/s10714-023-03130-7. arXiv: 2301.13017 [gr-qc].
- Curtright, Thomas and Hassan Alshal (Nov. 2022). “Newtonian Gravity on an N-Sphere”. In: *Bulg. J. Phys.* 50.1, pp. 1–12. arXiv: 2211.08236 [physics.class-ph].
- Ali, A. F., E. Moulay, K. Jusufi, and H. Alshal (2022). “Unitary symmetries in wormhole geometry and its thermodynamics”. In: *Eur. Phys. J. C* 82.12, p. 1170. DOI: 10.1140/epjc/s10052-022-11095-1. arXiv: 2302.08307 [hep-th].
- Hemeda, Mohammed, Hassan Alshal, Ahmed Farag Ali, and Elias C. Vagenas (Aug. 2022). “Gravitational Observations and LQGUP”. in: *Nucl. Phys. B* 1000, p. 116456. DOI: 10.1016/j.nuclphysb.2024.116456. arXiv: 2208.04686 [gr-qc].
- Danekar, Ashkbiz, Hassan Alshal, and Thomas L. Curtright (2021). “Dual Fields of Massive/Massless Gravitons in IR/UV Completions”. In: *Int. J. Mod. Phys. D* 30.14, p. 2142021. DOI: 10.1142/S0218271821420219. arXiv: 2109.05148 [hep-th].
- Van Kortryk, T. S., T. L. Curtright, and H. Alshal (2020). “On Enceladian Fields”. In: *Bulg. J. Phys.* 48.2, pp. 138–145. arXiv: 2012.13959 [physics.pop-ph].
- Vagenas, Elias C., Ahmed Farag Ali, Mohammed Hemeda, and Hassan Alshal (2020). “Massless Charged Particles Tunneling Radiation from a RN-dS Horizon and the Linear and Quadratic GUP”. in: *Ann. Phys.* 432, p. 168574. DOI: 10.1016/j.aop.2021.168574. arXiv: 2008.09853 [hep-th].
- Alshal, Hassan (2019). “Linearized Stability of Bardeen de-Sitter Thin-Shell Wormholes”. In: *EPL* 128. 6, p. 60007. DOI: 10.1209/0295-5075/128/60007. arXiv: 1909.07811 [gr-qc].
- Curtright, Thomas L., David B. Fairlie, and H. Alshal (2019). “A Galileon Primer”. In: arXiv: 1212.6972 [hep-th].
- Alshal, H. and T. L. Curtright (2019). “Massive Dual Gravity in N Spacetime Dimensions”. In: *JHEP* 09, p. 063. DOI: 10.1007/JHEP09(2019)063. arXiv: 1907.11537 [hep-th].
- Curtright, T. L. and H. Alshal (2019). “Massive Dual Spin 2 Revisited”. In: *Nucl. Phys. B* 948, p. 114777. DOI: 10.1016/j.nuclphysb.2019.114777. arXiv: 1907.11532 [hep-th].

- Vagenas, Elias C., Ahmed Farag Ali, and Hassan Alshal (2019). “Massless charged particles, naked singularity, and GUP in Reissner-Nordström-de Sitter-like spacetime”. In: *Phys. Rev. D* 99. 8, p. 084013. DOI: 10.1103/PhysRevD.99.084013. arXiv: 1903.09634 [hep-th].
- Vagenas, Elias C., Ahmed Farag Ali, Mohammed Hemeda, and Hassan Alshal (2019). “Linear and Quadratic GUP, Liouville Theorem, Cosmological Constant, and Brick Wall Entropy”. In: *Eur. Phys. J. C* 79. 5, p. 398. DOI: 10.1140/epjc/s10052-019-6908-z. arXiv: 1903.08494 [hep-th].
- Al-Modlej, Abeer, Salwa Alsaleh, Hassan Alshal, and Ahmed Farag Ali (2019). “Proton Decay and the Quantum Structure of Spacetime”. In: *Can. J. Phys.* 97, pp. 1317–1322. DOI: 10.1139/cjp-2018-0423. arXiv: 1903.02940 [hep-th].
- Vagenas, Elias C., Ahmed Farag Ali, and Hassan Alshal (2019). “GUP and the no-cloning theorem”. In: *Eur. Phys. J. C* 79. 3, p. 276. DOI: 10.1140/epjc/s10052-019-6789-1. arXiv: 1811.06614 [gr-qc].
- Alshal, H., T. Curtright, and S. Subedi (2018). “Image Charges Re-Imagined”. In: *Bulg. J. Phys.* 48.2, pp. 202–224. arXiv: 1808.08300 [physics.class-ph].
- Alshal, Hassan and Thomas Curtright (2018). “Grounded Hyperspheres as Squashed Wormholes”. In: *J. Math. Phys.* 60. 3, p. 032901. DOI: 10.1063/1.5044432. arXiv: 1806.03762 [physics.class-ph].
- Curtright, T., H. Alshal, P. Baral, S. Huang, J. Liu, K. Tamang, X. Zhang, and Y. Zhang (2018). “The Conducting Ring Viewed as a Wormhole”. In: *Eur. J. Phys.* 40. 1, p. 015206. DOI: 10.1088/1361-6404/aae3cd. arXiv: 1805.11147 [physics.class-ph].